



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester(2025-26)
Faculty Name	Ms. P. Swethanagasri
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Roll Number	2520090168
Branch	CSIT
Course Outcome	2

Case Study Topic: Flipkart product catalog — price-range scans dominate.

Work Done Summary:

Implemented efficient product catalog querying using a B+ Tree index on (category, price). The solution locates the starting key using logarithmic search and performs sequential leaf-page scans to retrieve all products within the specified price range. For an average query returning about 2,800 products, approximately 14 leaf pages are scanned (200 tuples per page). Analysis shows that the B+ Tree is optimal for the workload since 95% of queries are range scans, while a hash index would mainly benefit the small fraction of exact-match lookups.

Implementation Details :

1. Used a B+ Tree indexed on the composite key (category, price) to organize the product catalog.
2. Each leaf node stores up to 200 product tuples and maintains links to neighboring leaf nodes for efficient sequential access.
3. For a range query, the algorithm first locates the starting key using B+ Tree traversal and then scans consecutive leaf pages until the upper price limit is reached.
4. This approach minimizes disk I/O and efficiently handles large-scale range searches over 10^7 products stored on SSD.

Result : Screenshots/Output

```
Products in category = electronics with price in [12000.0, 14800.0]
```

```
(electronics, 12300.0, 102)
```

```
(electronics, 12900.0, 103)
```

```
(electronics, 13500.0, 104)
```

```
(electronics, 14100.0, 105)
```

```
(electronics, 14700.0, 106)
```

Time Complexity:

- **B+ Tree Construction:** $O(n)$
- **Range Query Search:** $O(\log n + k)$
- **Leaf Page Scan:** $O(k)$
- **Direct Product Lookup:** $O(\log n)$
- Where **n** is the total number of products and **k** is the number of products returned in the range query

GitHub Link :-

Student Signature	Faculty Signature